

Famitinib for familial adenomatous polyposis-associated aggressive desmoid tumors: 32-month follow-up from a single-center exploratory study.

Sai Ge, Jian Li, Xicheng Wang, Bo Zhuang, Changhong Zhao, Lin Shen; Department of Gastrointestinal Oncology, Key laboratory of Carcinogenesis and Translational Research (Ministry of Education), Peking University Cancer Hospital & Institute, Beijing, China; Jiangsu Hengrui Pharmaceuticals Co., Ltd., Shanghai, China; Department of Gastrointestinal Oncology, Key Laboratory of Carcinogenesis and Translational Research (Ministry of Education), Peking University Cancer Hospital & Institute, Beijing, China

Background: Desmoid tumors (DTs) associated with familial adenomatous polyposis (FAP) are rare, locally aggressive soft-tissue tumors with high recurrence rates and poor prognosis. Famitinib, an orally administered multi-targeted tyrosine kinase inhibitor targeting VEGFR-2, PDGFR, c-KIT, and FGFR, has shown promising efficacy in a prospective, single-arm study for these patients (pts). This report presents updated efficacy and safety data from a 32-month follow-up. **Methods:** Pts with FAP carrying germline APC mutations and pathologically confirmed DTs that progressed within 6 months according to RECIST v1.1 criteria were enrolled. Famitinib was administered at 20 mg once daily in 3-week cycles. The primary endpoint was the objective response rate (ORR). Secondary endpoints included disease control rate (DCR), progression-free survival (PFS), overall survival (OS), and safety. **Results:** Between November 2021 and March 2023, 12 eligible pts were enrolled, with a median age of 35.5 years (range: 25–53). Of these, 41.7% (5/12) were male. Intra-abdominal (IA) DTs were present in 83.3% (10/12) of pts, while 2 pts had extra-abdominal (EA) DTs. Eight pts presented with stage III–IV DTs characterized by rapid progression and severe symptoms. As of January 20, 2025, the median follow-up duration was 32.2 months (21.8–37.9). Seven pts achieved partial response (PR), and five achieved stable disease (SD), with a median time to response of 7.1 months (4.1–11.7). The confirmed ORR was 50%, with a DCR of 100%. Among pts with IA DTs, the confirmed ORR was 60.0% (6/10). One patient who achieved PR withdrew from the study after being diagnosed with duodenal cancer. The 6-month and 1-year PFS rates were 100% and 91.7%, respectively, while the 1-year OS rate was 100%. At 2 years, the PFS and OS rates were 54.5% and 72.7%, respectively. The median PFS and OS were not reached. Treatment-emergent adverse events (TEAEs) occurred in all pts (100%). The most common TEAEs (all grades) included COVID-19 (91.7%), leukopenia (83.3%), hypertension (83.3%), neutropenia (83.3%), proteinuria (66.7%), and elevated bilirubin levels (50%). Grade 3 TEAEs occurred in six pts, including neutropenia (41.7%), leukopenia (25%), hypertension (16.7%), hand-foot syndrome (8.3%), intestinal obstruction (8.3%), and abdominal hemorrhage (8.3%). One patient experienced a grade 4 adverse event (intestinal perforation), declined surgical intervention, and passed away three months later. All male pts tolerated the 20 mg dose, while 5 female pts (71.4%) reduced to 15 mg. **Conclusions:** Famitinib demonstrated sustained clinical efficacy and meaningful survival benefits in pts with FAP-associated aggressive DTs after a median follow-up of 32 months. Given the unique characteristics, careful monitoring for intestinal perforation and the risk of second primary tumors is crucial during treatment. Clinical trial information: ChiCTR2100051307. Research Sponsor: None.